Google Ads MCP Tool

Design Documentation

Version 2.0 - March 2026

1. Overview

Google Ads MCP Tool is an internal campaign management and reporting application built to help our digital marketing agency (Awen Studio) manage Google Ads campaigns for our clients. The tool connects to the Google Ads API to create, manage, and optimize campaigns, retrieve campaign data, generate performance reports, and monitor account metrics in real time.

The tool is used exclusively by our internal team and is not distributed or sold to third parties.

2. Key Features

- List and manage accessible Google Ads accounts via MCC hierarchy
- Create, update, pause, and remove campaigns across all campaign types
- Manage ad groups, ads, keywords, bids, and budgets programmatically
- Execute GAQL (Google Ads Query Language) queries for custom reporting
- Retrieve campaign performance metrics (impressions, clicks, cost, conversions, CTR, CPC)
- Support for all campaign types: Search, Display, Shopping, Video, Performance Max, App, Demand Gen
- Monthly and daily performance history tracking
- Multi-account management through MCC hierarchy

3. Architecture

The application follows a client-server architecture using the Model Context Protocol (MCP):

- **MCP Server:** A Python-based server that exposes tools for interacting with the Google Ads API.
- **Google Ads API Client:** Uses the official google-ads-python library (v23) to communicate with Google Ads API endpoints.
- **Authentication:** OAuth 2.0 with refresh token for secure, persistent access.
- **Data Layer:** GAQL queries are executed via GoogleAdsService.SearchStream for efficient data retrieval.
- **Management Layer:** Campaign, ad group, ad, and keyword mutations are executed via the corresponding Google Ads API mutate services.

4. Google Ads API Usage

The tool uses the following Google Ads API services:

Read Operations (Reporting):

- **CustomerService.ListAccessibleCustomers:** To list accounts the user has access to.
- **GoogleAdsService.SearchStream:** To execute GAQL queries and retrieve reporting data.

Write Operations (Campaign Management):

- **CampaignService.MutateCampaigns:** To create, update, pause, and remove campaigns.
- **AdGroupService.MutateAdGroups:** To manage ad groups within campaigns.
- **AdGroupAdService.MutateAdGroupAds:** To create and manage ads.
- **AdGroupCriterionService.MutateAdGroupCriteria:** To manage keywords and targeting criteria.
- **CampaignBudgetService.MutateCampaignBudgets:** To create and adjust campaign budgets.

The tool provides both read and write capabilities. All write operations include validation, error handling, and confirmation workflows to prevent unintended changes.

Rate limiting and error handling follow Google Ads API best practices, including proper handling of GoogleAdsException errors.

5. Authentication & Security

- OAuth 2.0 authentication flow with refresh token.
- Credentials are stored securely in local configuration files (google-ads.yaml).
- The tool supports login_customer_id for MCC-level access control.
- No user credentials are transmitted to or stored on external servers.
- The application runs locally and is not exposed to the public internet.
- All mutate operations require explicit confirmation before execution.

6. Data Handling & Privacy

- All data retrieved from the Google Ads API is used solely for internal campaign management and reporting.
- No personal user data is collected, stored, or shared with third parties.
- Campaign performance data is processed in memory and displayed to authorized team members only.
- The tool complies with Google Ads API Terms of Service and data usage policies.

7. Technology Stack

- Python 3.12
- google-ads Python client library (v23)
- FastMCP (Model Context Protocol server framework)
- OAuth 2.0 for authentication
- Protocol Buffers (protobuf) for API communication